class08

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Preparing the data

```
# Importing and naming data set
fna.data <- "WisconsinCancer.csv"
wisc.df <- read.csv(fna.data, row.names=1)

# removing the diagnosis column
wisc.data <- wisc.df[,-1]

# creating diagnosis vector for later
diagnosis <- wisc.df$diagnosis
```

Exploratory data analysis

Q1: How many observations are in this dataset?

31 observations in this dataset

```
# how many observations in wisc.df?
length(wisc.df)
```

```
[1] 31
```

Q2: How many of the observations have a malignant diagnosis?

212 observations have a malignant diagnosis

```r
# Seeing how many of each diagnosis in the dataset
table(wisc.df$diagnosis)
```

```

  B   M 
357 212 
```

Q3: How many variables/features in the data are suffixed with _mean?

10 variables are suffixed with "_mean"

```r
length(grep("_mean", names(wisc.df)))
```

```
[1] 10
```

Performing PCA

```r
# Checking column means
colMeans(wisc.data)
```

```
            radius_mean            texture_mean          perimeter_mean 
           1.412729e+01            1.928965e+01            9.196903e+01 
              area_mean         smoothness_mean        compactness_mean 
           6.548891e+02            9.636028e-02            1.043410e-01 
         concavity_mean     concave.points_mean           symmetry_mean 
           8.879932e-02            4.891915e-02            1.811619e-01 
 fractal_dimension_mean               radius_se              texture_se 
           6.279761e-02            4.051721e-01            1.216853e+00 
           perimeter_se                 area_se           smoothness_se 
           2.866059e+00            4.033708e+01            7.040979e-03 
         compactness_se            concavity_se       concave.points_se 
           2.547814e-02            3.189372e-02            1.179614e-02 
            symmetry_se    fractal_dimension_se            radius_worst 
           2.054230e-02            3.794904e-03            1.626919e+01 
          texture_worst         perimeter_worst              area_worst 
           2.567722e+01            1.072612e+02            8.805831e+02 
       smoothness_worst       compactness_worst         concavity_worst 
           1.323686e-01            2.542650e-01            2.721885e-01 
   concave.points_worst          symmetry_worst fractal_dimension_worst 
           1.146062e-01            2.900756e-01            8.394582e-02 
```

```r
# Checking standard deviations
apply(wisc.data, 2, sd)
```

```
            radius_mean            texture_mean          perimeter_mean 
           3.524049e+00            4.301036e+00            2.429898e+01 
              area_mean         smoothness_mean        compactness_mean 
           3.519141e+02            1.406413e-02            5.281276e-02 
         concavity_mean     concave.points_mean           symmetry_mean 
           7.971981e-02            3.880284e-02            2.741428e-02 
 fractal_dimension_mean               radius_se              texture_se 
           7.060363e-03            2.773127e-01            5.516484e-01 
           perimeter_se                 area_se           smoothness_se 
           2.021855e+00            4.549101e+01            3.002518e-03 
         compactness_se            concavity_se       concave.points_se 
           1.790818e-02            3.018606e-02            6.170285e-03 
            symmetry_se    fractal_dimension_se            radius_worst 
           8.266372e-03            2.646071e-03            4.833242e+00 
          texture_worst         perimeter_worst              area_worst 
           6.146258e+00            3.360254e+01            5.693570e+02 
       smoothness_worst       compactness_worst         concavity_worst 
           2.283243e-02            1.573365e-01            2.086243e-01 
   concave.points_worst          symmetry_worst fractal_dimension_worst 
           6.573234e-02            6.186747e-02            1.806127e-02 
```

```r
# Performing PCA on wisc.data
wisc.pr <- prcomp(wisc.data, scale = TRUE)

# Summary of results
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
```

```
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

Q4: From your results, what proportion of the original variance is captured by the first principal component (PC1)?

0.4427 or 44.27% of original variance.

Q5: How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

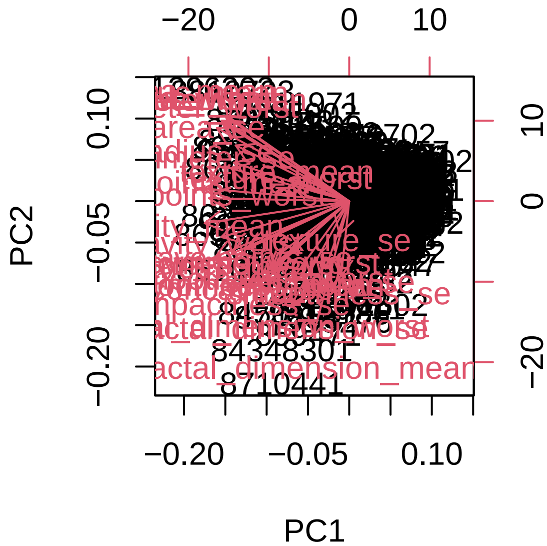
Three PCs are required to describe at least 70% of the original variance.

Q6: How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

7 PCs are required to describe at least 90% of the original variance.

Interpreting PCA results

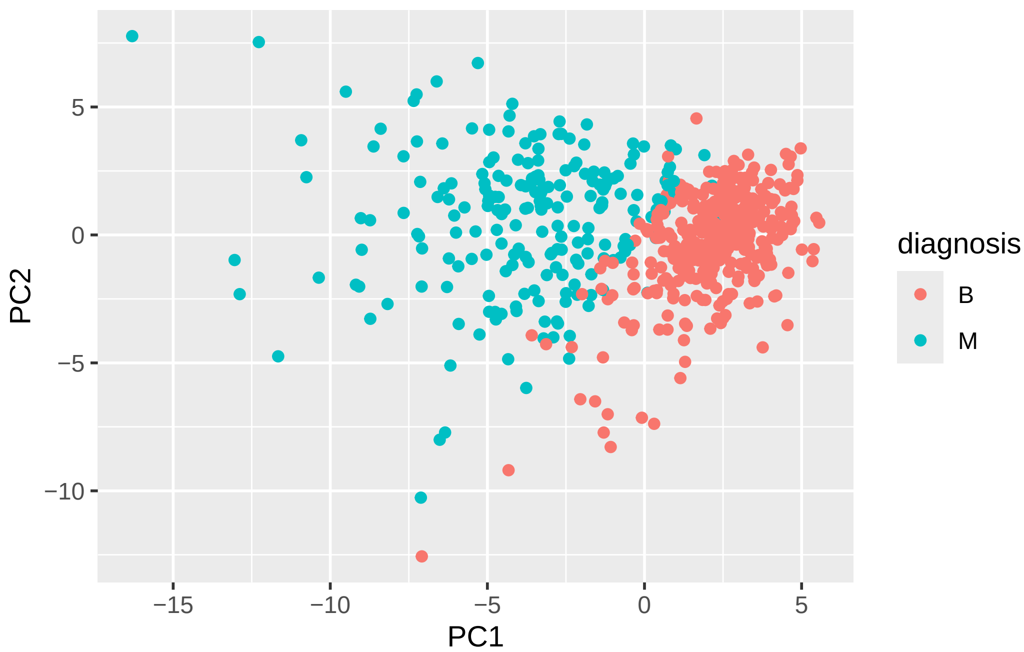
```r
biplot(wisc.pr)
```



Q7: What stands out to you about this plot? Is it easy or difficult to understand? Why?

The plot is really clustered together and, because of this, it is difficult to understand.

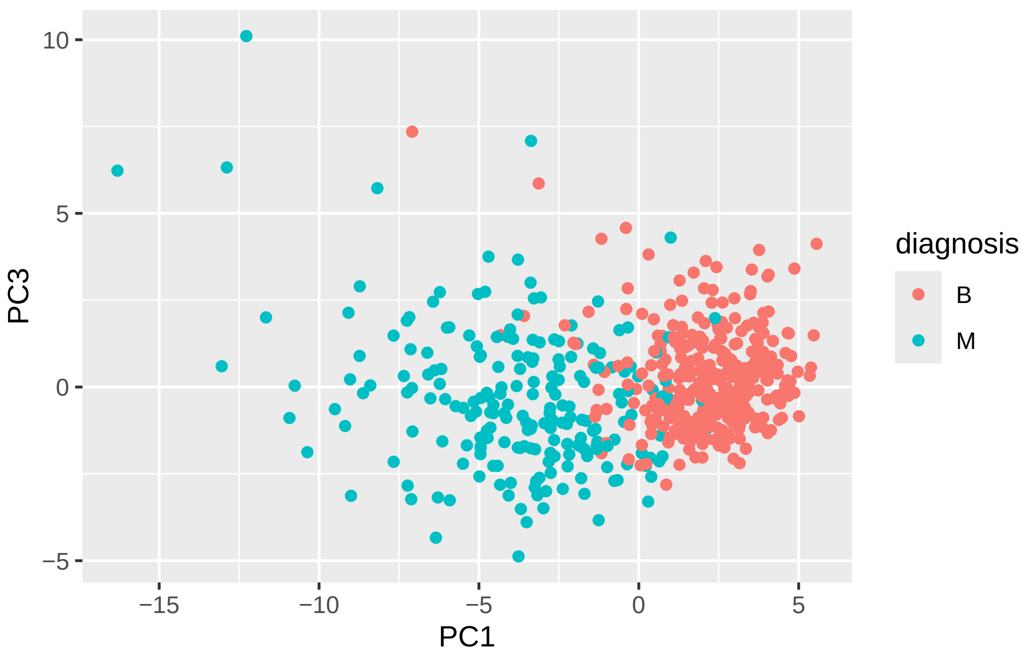
```r
library(ggplot2)

# Scatter plot observations by components 1 and 2
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Q8: Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

The PC1 vs. PC3 plot has more separation between malignant and benign tumors than the PC1 vs. PC2 plot.

```
# Scatter plot observations by components 1 and 3
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```

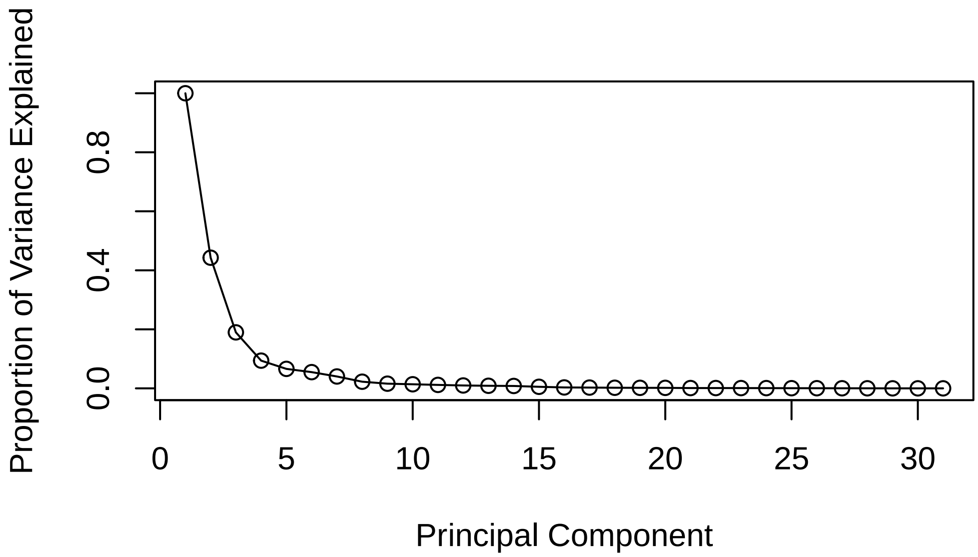


Variance explained

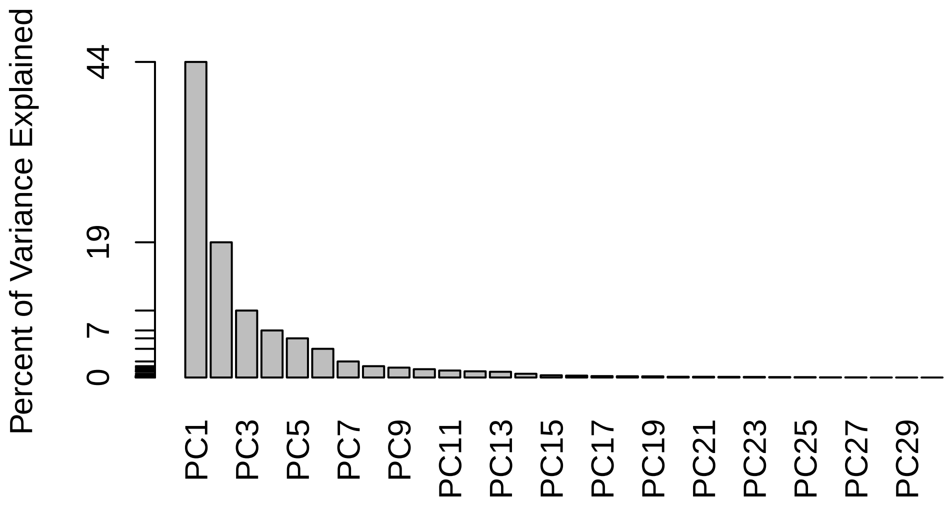
```
# Calculating variance of each PC
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
# Calculating variance explain by each PC
pve <- pr.var/sum(pr.var)

# Plot pve
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



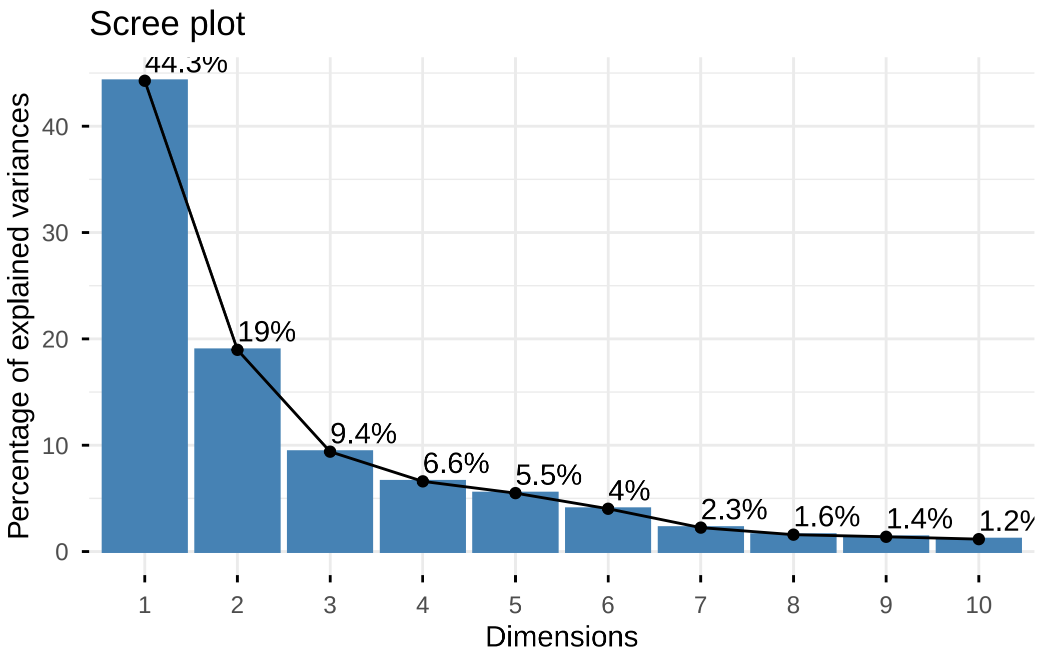

```
# Alternative scree plot of the same data
barplot(pve, ylab = "Percent of Variance Explained",
     names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```




```
# Additional alternative plot using the same data
library(factoextra)
```

```
Welcome to factoextra!

Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/
guide-to-principal-component-methods-in-r/
```

```
fviz_eig(wisc.pr, addlabels = TRUE)
```



Communicating PCA results

Q9: For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

-0.26085376. This is the feature with the largest contribution.

```
# Feature contributions to PC1
wisc.pr$rotation[,1]
```

```
            radius_mean            texture_mean          perimeter_mean
            -0.21890244             -0.10372458             -0.22753729
              area_mean         smoothness_mean        compactness_mean
            -0.22099499             -0.14258969             -0.23928535
         concavity_mean     concave.points_mean           symmetry_mean
            -0.25840048             -0.26085376             -0.13816696
 fractal_dimension_mean               radius_se              texture_se
            -0.06436335             -0.20597878             -0.01742803
           perimeter_se                 area_se           smoothness_se
```

```
          -0.21132592             -0.20286964             -0.01453145
       compactness_se            concavity_se       concave.points_se
          -0.17039345             -0.15358979             -0.18341740
          symmetry_se    fractal_dimension_se            radius_worst
          -0.04249842             -0.10256832             -0.22799663
        texture_worst         perimeter_worst              area_worst
          -0.10446933             -0.23663968             -0.22487053
     smoothness_worst       compactness_worst         concavity_worst
          -0.12795256             -0.21009588             -0.22876753
 concave.points_worst          symmetry_worst fractal_dimension_worst
          -0.25088597             -0.12290456             -0.13178394
```

Hierarchical clustering

```r
# Scaling data
data.scaled <- scale(wisc.data)
```

```r
# Calculating distances between pairs of observations
data.dist <- dist(data.scaled)
```

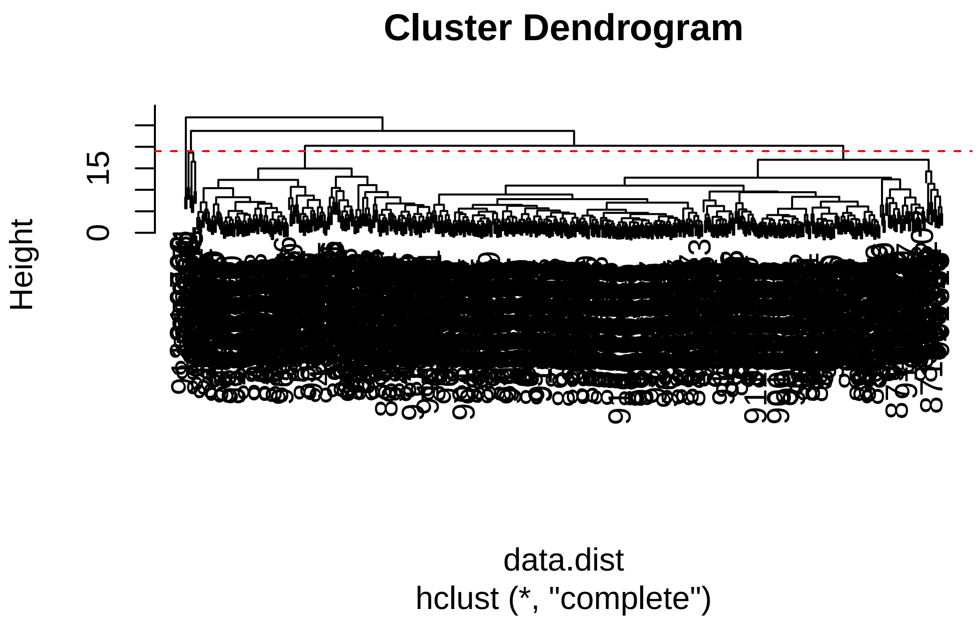
```r
# Creating hierarchical clustering model using complete linkage
wisc.hclust <- hclust(data.dist, method = "complete")
```

Results of hierarchical clustering

Q10: Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

The height at which the clustering model has 4 clusters is 19.

```r
plot(wisc.hclust)
abline(h=19, col="red", lty=2)
```



Selecting number of clusters

```
# Cutting the tree to have 4 clusters
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)

# Comparing cluster membership to actual diagnosis
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

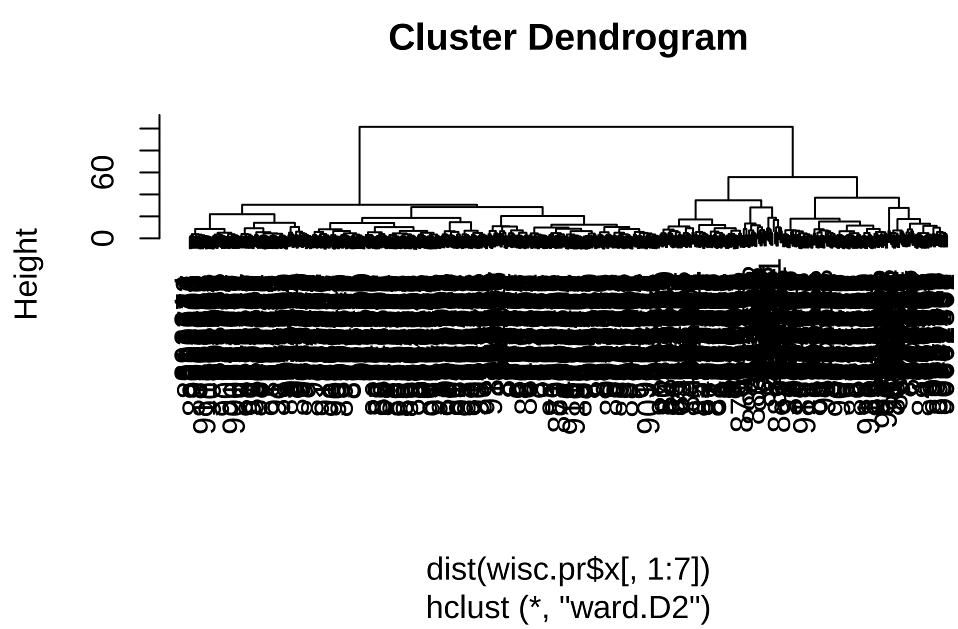
Using different methods

Q12: Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

My favorite method is "complete" because it creates less significant clusters. Although it does not lead to as much specificity of diagnosis within each cluster compared to "ward.D2," I like the simplicity of less clusters even if there is a little more variance.

Clustering of PCA results

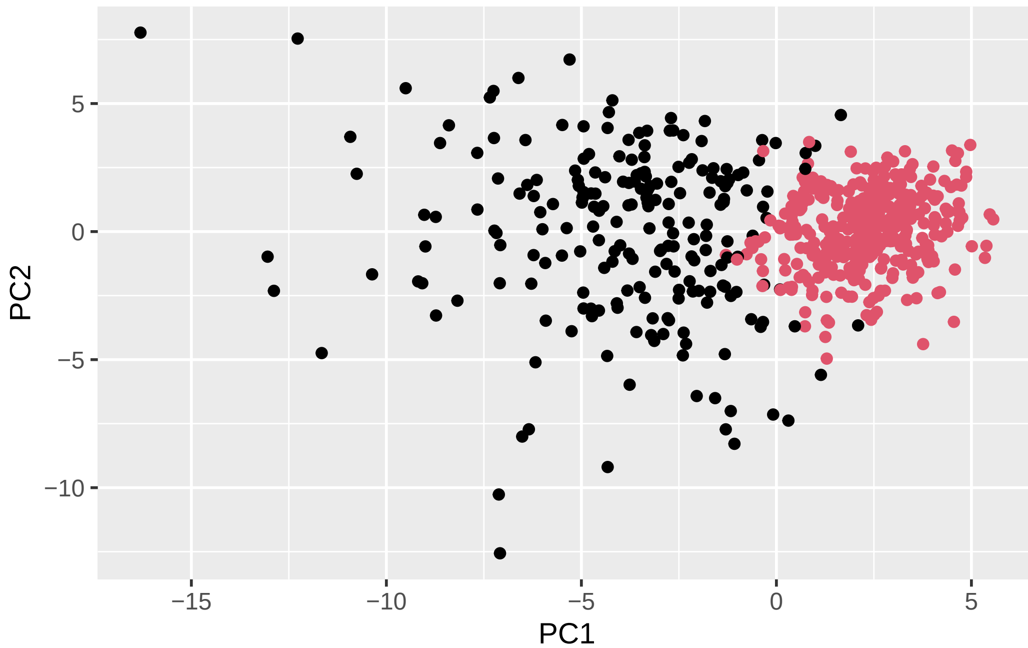
```
# Using Ward's criterion for clustering along the first 7 PCs
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")

# Plotting the hierarchical clustering model
plot(wisc.pr.hclust)
```




```
# Creating two cluster groups
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)
table(wisc.pr.hclust.clusters)
```

```
wisc.pr.hclust.clusters
  1   2
216 353
```

```
# Plotting the clusters
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=wisc.pr.hclust.clusters)
```




```
# Comparing to actual diagnoses
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

Q13: How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

The malignant and benign diagnoses are separated well; better in PC2 than in PC1.

Q14: How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses?

The initial clustering model appears to be better at separating the diagnoses, as shown below. Although this is true, there are still clusters with few data points, effectively creating two "real" clusters.

```r
table(wisc.hclust.clusters, diagnosis)
```

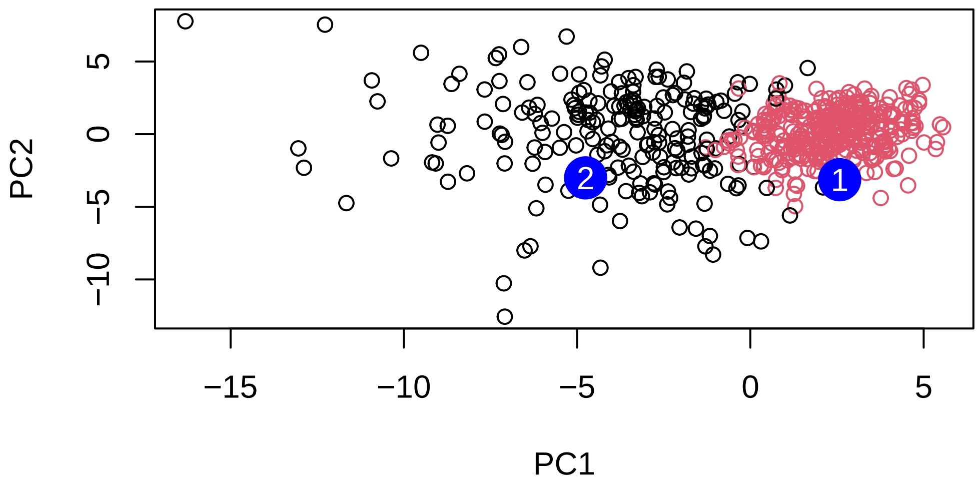
```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

Prediction

```r
# url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```r
# Plotting the new data
plot(wisc.pr$x[,1:2], col=wisc.pr.hclust.clusters)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16: Which of these new patients should we prioritize for follow up based on your results?

Patient 1 should be prioritized for follow-up.